Skill Builder Pack 4B — SUM / SUMIF / SUMIFS

We will now use SKILL 4B SUMS

Key Idea

We are now answering:

> “How much?”

Instead of counting rows, Excel will:

- look down a column
- take the numbers
- add them together

SECTION 1 — SUM (no conditions)

Question 1

What is the total engagement score?

What you do:

Use SUM on the Engagement Score column

What Excel is doing:

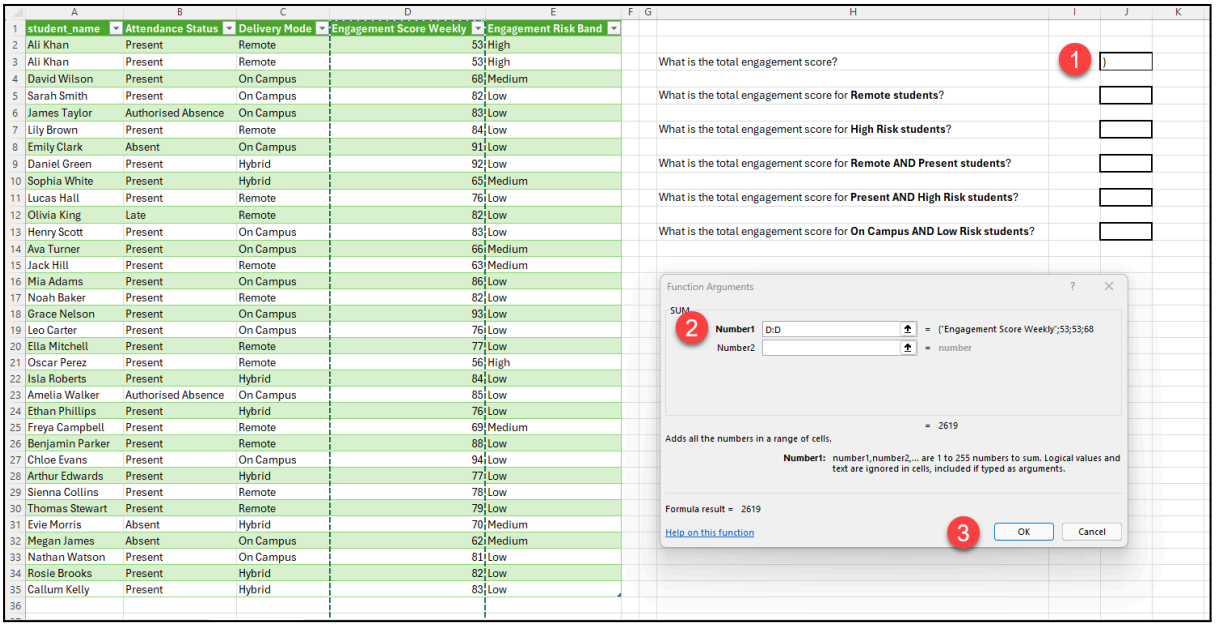
> Excel looks down the Engagement Score column.
> It takes each number and adds it to a running total.

So:

- Row 1 → total = 85
- Row 2 → total = 85 + 45 = 130
- Row 3 → total = 130 + 70 = 200

It keeps going until the end

1. Click on cell J3, open fx and search for SUM, then select it in the list and click OK to open the Wizard
2. Click in the **Number1** box, then select the **Engagement Score Weekly** column
3. Click OK



Final result:

👉 Total of all scores should be 2, 619

SECTION 2 — SUMIF (one condition)

Key Idea

SUMIF = add numbers where ONE rule is true

Question 2

What is the total engagement score for **Remote students**?

What Excel is doing:

1. Excel looks down each row.
2. For each row, it checks:
3. “Is Delivery Mode = remote?”

If YES → add the score
If NO → ignore it

Exampl

e:

Delivery	Score	What happens
Remote	85	Add it
Online	45	Ignore
On Campus	70	Ignore

Only Remote scores are added

1. Click on cell J5, select fx and then search for SUMIF and click on OK to open the Wizard
2. For **Range**, select the column we want to filter by, in this case Delivery Mode (Column C)
3. For Criteria, type in the value we want returned (*Remote*)
4. For **Sum_range** select the column with the numbers we want to add (in this case *Engagement Score Weekly* which is column D)
5. Press OK - you should get the value 940 back

student_name	Attendance Status	Delivery Mode	Engagement Score Weekly	Engagement Risk Band
Ali Khan	Present	Remote	53	High
Ali Khan	Present	Remote	53	High
David Wilson	Present	On Campus	68	Medium
Sarah Smith	Present	On Campus	82	Low
James Taylor	Authorised Absence	On Campus	83	Low
Lily Brown	Present	Remote	84	Low
Emily Clark	Absent	On Campus	91	Low
Daniel Green	Present	Hybrid	92	Low
Sophia White	Present	Hybrid	65	Medium
Lucas Hall	Present	Remote	76	Low
Olivia King	Late	Remote	82	Low
Henry Scott	Present	On Campus	83	Low
Ava Turner	Present	On Campus	66	Medium
Jack Hill	Present	Remote	63	Medium
Mia Adams	Present	On Campus	86	Low
Noah Baker	Present	Remote	82	Low
Grace Nelson	Present	On Campus	93	Low
Leo Carter	Present	On Campus	76	Low
Ella Mitchell	Present	Remote	77	Low
Oscar Perez	Present	Remote	56	High
Isla Roberts	Present	Hybrid	84	Low
Amelia Walker	Authorised Absence	On Campus	85	Low
Ethan Phillips	Present	Hybrid	76	Low
Freya Campbell	Present	Remote	69	Medium
Benjamin Parker	Present	Remote	88	Low
Chloe Evans	Present	On Campus	94	Low
Arthur Edwards	Present	Hybrid	77	Low
Sienna Collins	Present	Remote	78	Low
Thomas Stewart	Present	Remote	79	Low
Evie Morris	Absent	Hybrid	70	Medium
Megan James	Absent	On Campus	62	Medium
Nathan Watson	Present	On Campus	81	Low
Rosie Brooks	Present	Hybrid	82	Low
Callum Kelly	Present	Hybrid	83	Low

What is the total engagement score?	2619
What is the total engagement score for **Remote students**?	e",D:D)
What is the total engagement score for **High Risk students**?	
What is the total engagement score for **Remote AND Present students**?	
What is the total engagement score for **Present AND High Risk students**?	
What is the total engagement score for **On Campus AND Low Risk students**?	

Function Arguments

SUMIF

Range C:C = {"Delivery Mode";"Remote";"Remote";"

Criteria "Remote" = "Remote"

Sum_range D:D = {"Engagement Score Weekly";53;53;68

= 940

Adds the cells specified by a given condition or criteria.

Sum_range are the actual cells to sum. If omitted, the cells in range are used.

Formula result = 940

Help on this function

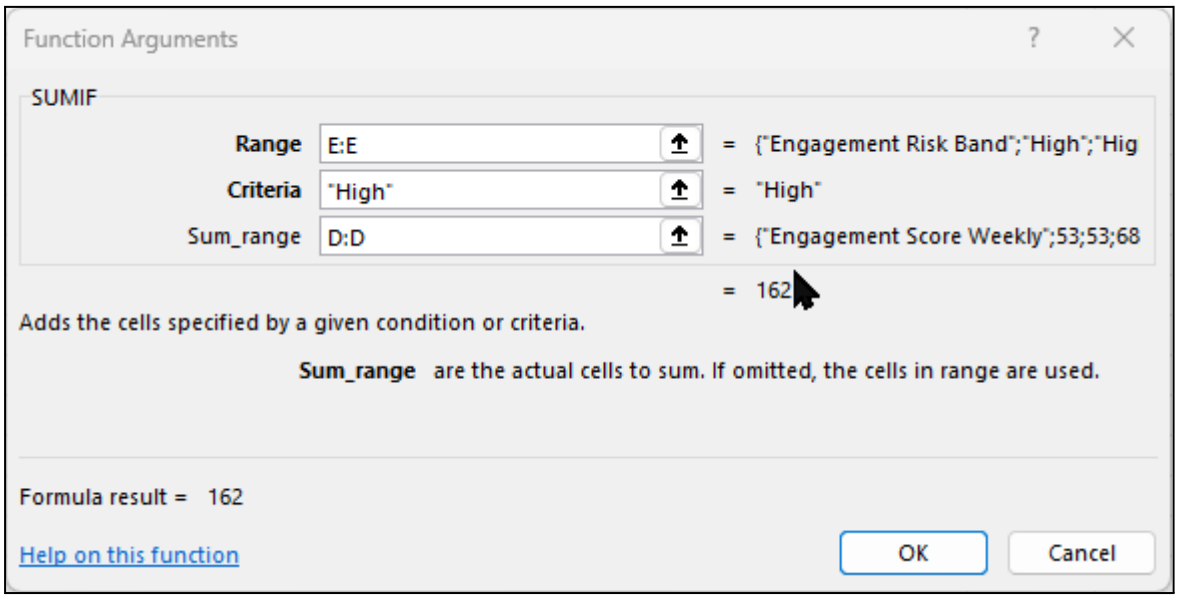
OK Cancel

Question 3

What is the total engagement score for **High Risk students**?

What Excel is doing:

1. Excel checks the Risk column
2. If it sees “High” → it adds the score
3. Otherwise → ignore



You should get **162** back as the sum value

SECTION 3 — SUMIFS (multiple conditions)

Key Idea

SUMIFS = add numbers where MULTIPLE rules are true

> "this AND this"

Question 4

What is the total engagement score for **Remote AND Present students**?

What Excel is doing:

For each row:

1. Is Delivery Mode = Remote?
2. Is Attendance = Present?

Only if BOTH are true → add the score

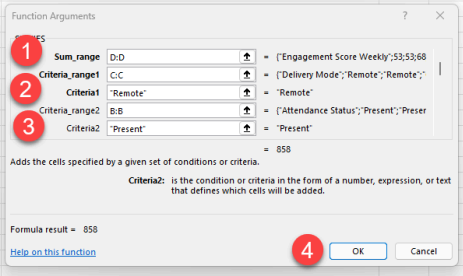
Example:

Delivery	Attendance	Score	Add it?
Remote	Present	85	✅ Yes
Online	Absent	45	❌ No
In Person	Present	70	❌ No

1. Select cell J9, select fx and then search for SUMIFS. Click on OK to open the Wizard
2. For **Sum_range**, select the numbers we want to add together, in this case *Engagement Score Weekly* (Column D)
3. For **Criteria_range1**, select *Delivery Mode (Column C)* and for **Criteria1**, type in *Remote*
4. For **Criteria_range2**, select *Attendance Status (Column B)* and for the **Criteria2**, type in *Present*
5. Click OK - you should get the value 858 back

student_name	Attendance Status	Delivery Mode	Engagement Score Weekly	Engagement Risk Band
Ali Khan	Present	Remote	53	High
Ali Khan	Present	Remote	53	High
David Wilson	Present	On Campus	68	Medium
Sarah Smith	Present	On Campus	82	Low
James Taylor	Authorised Absence	On Campus	83	Low
Lily Brown	Present	Remote	84	Low
Emily Clark	Absent	On Campus	91	Low
Daniel Green	Present	Hybrid	92	Low
Sophia White	Present	Hybrid	65	Medium
Lucas Hall	Present	Remote	76	Low
Olivia King	Late	Remote	82	Low
Henry Scott	Present	On Campus	83	Low
Ava Turner	Present	On Campus	66	Medium
Jack Hill	Present	Remote	63	Medium
Mia Adams	Present	On Campus	86	Low
Noah Baker	Present	Remote	82	Low
Grace Nelson	Present	On Campus	93	Low
Leo Carter	Present	On Campus	76	Low
Ella Mitchell	Present	Remote	77	Low
Oscar Perez	Present	Remote	56	High
Isla Roberts	Present	Hybrid	84	Low
Amelia Walker	Authorised Absence	On Campus	85	Low
Ethan Phillips	Present	Hybrid	76	Low
Freya Campbell	Present	Remote	69	Medium
Benjamin Parker	Present	Remote	88	Low
Chloe Evans	Present	On Campus	94	Low
Arthur Edwards	Present	Hybrid	77	Low
Sienna Collins	Present	Remote	78	Low
Thomas Stewart	Present	Remote	79	Low
Evie Morris	Absent	Hybrid	70	Medium
Megan James	Absent	On Campus	62	Medium
Nathan Watson	Present	On Campus	81	Low
Rosie Brooks	Present	Hybrid	82	Low
Callum Kelly	Present	Hybrid	83	Low

What is the total engagement score?	2619
What is the total engagement score for **Remote students**?	940
What is the total engagement score for **High Risk students**?	162
What is the total engagement score for **Remote AND Present students**?	ent")
What is the total engagement score for **Present AND High Risk students**?	
What is the total engagement score for **On Campus AND Low Risk students**?	

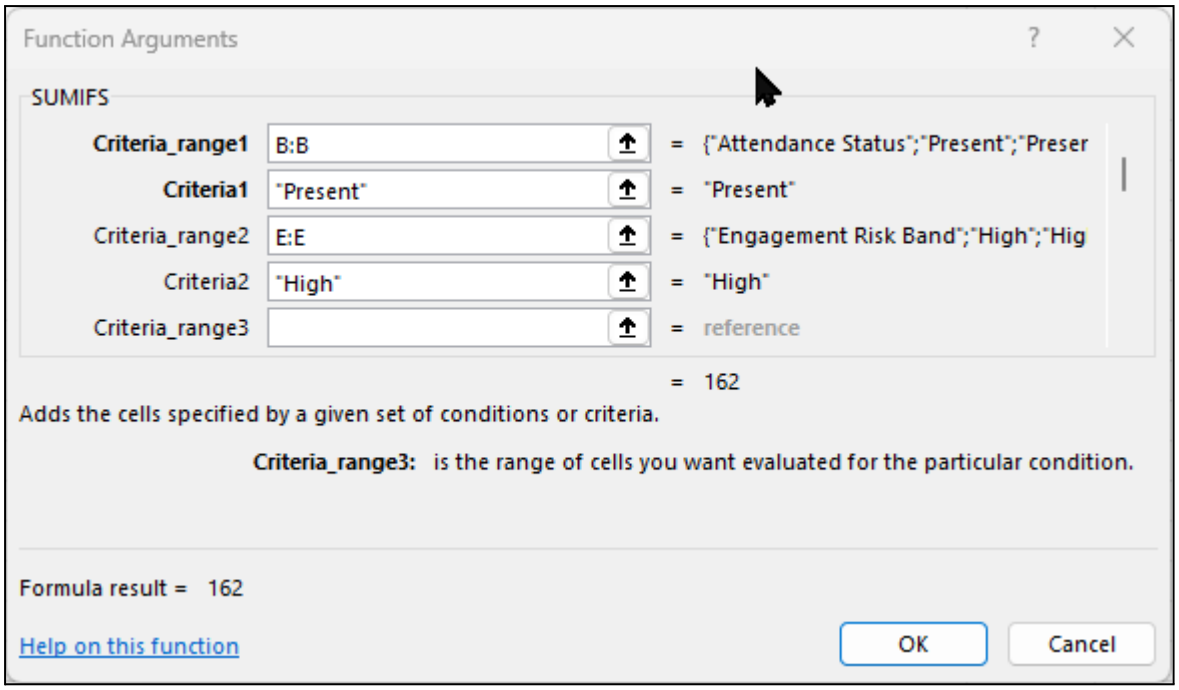


Question 5

What is the total engagement score for **Present AND High Risk students**?

What Excel is doing:

- Excel checks BOTH conditions
- It only adds the score if BOTH match



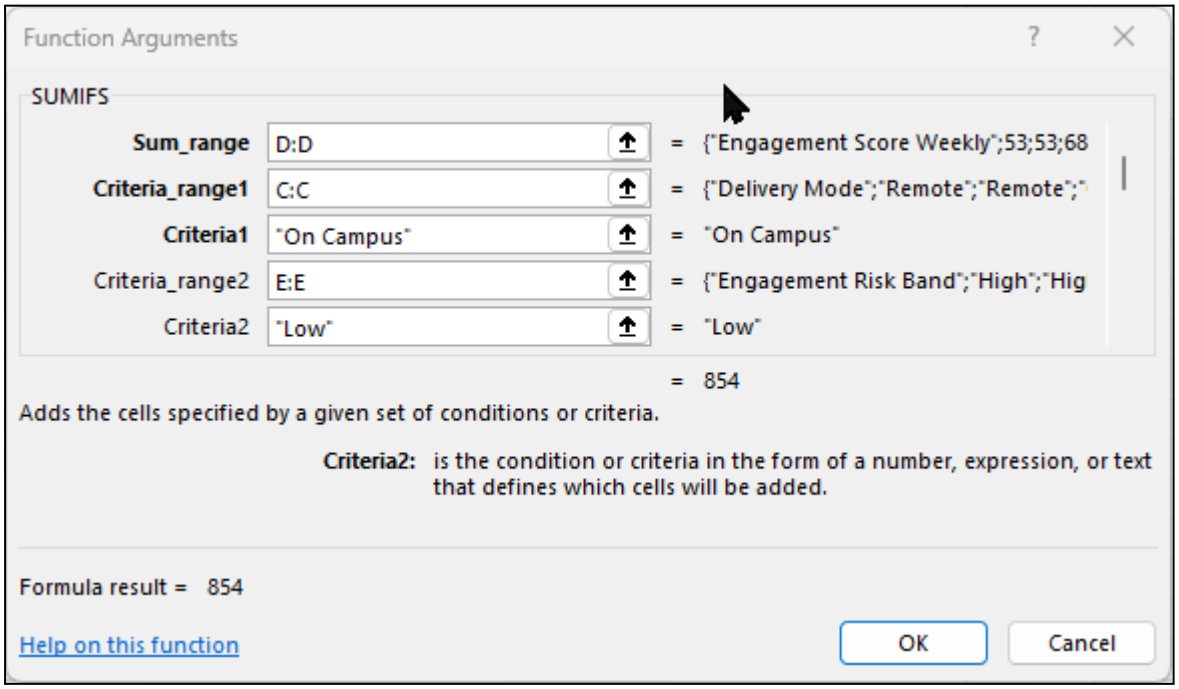
You should get the result **162**

Question 6

What is the total engagement score for **On Campus AND Low Risk students**?

Same logic again:

- Check row → both rules true → add number
- Otherwise → ignore



You should get the result **854**

Final Summary

What Excel is doing every time:

- It looks at one row at a time
- It checks your rule(s)
- If rules match → it adds the number
- If not → it skips it

The only difference between functions:

- SUM → add everything
- SUMIF → add if ONE rule is true
- SUMIFS → add if MULTIPLE rules are true

The simplest way to remember:

1. SUM = “Add everything”
2. SUMIF = “Add numbers that match this”
3. SUMIFS = “Add numbers that match this AND this”

Final note

If you get stuck:

> “Excel is just going row by row and deciding whether to add the number or ignore it”